Business Calculus (MATH 2350)
Summer 2019
Tuesday, 06/04/2019
Quiz 2: Section 10.5,10.7

Name (Print): ____________________

Score: / 100

Section 10.5

1. Find the derivative $g'(t)$ of the given function $g(t)$.

   (a) $\boxed{\text{10 points}}$ $g(t) = \dfrac{4}{\sqrt{t}}$

   (b) $\boxed{\text{10 points}}$ $g(t) = 4 + 4t + 4t^2 + 4t + 4$

   (c) $\boxed{\text{10 points}}$ $g(t) = (t+1)^3 + 9$

2. $\boxed{\text{10 points}}$ Let f and g be functions satisfying $f'(3) = 7$ and $g'(3) = 1$.
If $h(x) = f(x) - 2g(x) + 2$, find $h'(3)$.

3. $\boxed{\text{10 points}}$ A company's cost (in dollars) for producing x desks is given by

$$C(x) = 100x - 10x^2 + x^3.$$

Find $C(5)$ and $C'(5)$. Show your work. Write a brief, informal interpretation of your results.

4. $\boxed{\text{10 points (bonus)}}$ Let $f(x) = x^4$ and let $g(x) = x^2 + 2$. Using the fact that the correct derivative of $f(g(x))$ is given by

$$(f(g(x)))' = 8x(x^2 + 2)^3,$$

which one of the following statements is true?

(a) $(f(g(x)))' = f(g'(x))f'(x)$
(b) $(f(g(x)))' = f'(g(x))g'(x)$
(c) $(f(g(x)))' = f'(x)g'(x)$

Hint: You don't need to expand $8x(x^2 + 2)^3$ *to get the correct answer.*

Section 10.7

1. The price-demand equation for the production of bicycles is given by the following equation, where x is the quantity of bicycles demanded and p is price (in dollars)

$$x = 5000 - 25p.$$

(a) $\boxed{\text{15 points}}$ Express the revenue $R(x)$ as a function of the demand x.
Hint: revenue equals price times demand.

(b) $\boxed{\text{10 points}}$ Find the marginal revenue $R'(x)$.

(c) $\boxed{\text{10 points}}$ Find $R(2500)$ and $R'(2500)$, and give a brief, informal interpretation of your results.

2. $\boxed{\text{15 points}}$ If the cost of producing x cell phones is given by $C(x) = 300x + 100$, find the marginal average cost of producing x phones.